

Low Titer Group O Whole Blood In Injured Children Requiring Massive Transfusion

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Objective: The aim of this study was to assess the survival impact of low-titer group O whole blood (LTOWB) in injured pediatric patients who require massive transfusion.

Summary Background Data: Limited data are available regarding the effectiveness of LTOWB in pediatric trauma.

Methods: A prospective observational study of children requiring massive transfusion after injury at UPMC Children's Hospital of Pittsburgh, an urban academic pediatric Level 1 trauma center. Injured children ages 1 to 17 years who received a total of >40 mL/kg of LTOWB and/or conventional components over the 24 hours after admission were included. Patient characteristics, blood product utilization and clinical outcomes were analyzed using Kaplan-Meier survival curves, log rank tests and Cox proportional hazards regression analyses. The primary outcome was 28-day survival.

Results: Of patients analyzed, 27 of 80 (33%) received LTOWB as part of their hemostatic resuscitation. The LTOWB group was comparable to the component therapy group on baseline demographic and physiologic parameters except older age, higher body weight, and lower red blood cell and plasma transfusion volumes. After adjusting for age, total blood product volume transfused in 24 hours, admission base deficit, international normalized ratio (INR), and injury severity score (ISS), children who received LTOWB as part of their resuscitation had significantly improved survival at both 72 hours and 28 days post-trauma [adjusted odds ratio (AOR) 0.23, $P = 0.009$ and AOR 0.41, $P = 0.02$, respectively]; 6-hour survival was not statistically significant (AOR = 0.51, $P = 0.30$). Survivors at 28 days in the LTOWB group had reduced hospital LOS, ICU LOS, and ventilator days compared to the CT group.

Conclusion: Administration of LTOWB during the hemostatic resuscitation of injured children requiring massive transfusion was independently associated with improved 72-hour and 28-day survival.

Keywords: child, hemorrhage, massive transfusion, trauma, whole blood
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The following conflicts of interest are relevant to this study: M.H.Y. has given paid lectures and received travel reimbursement from Terumo BCT, the manufacturer of the PLT-sparing leukoreduction filter used in the preparation of the LTOWB units.

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Trauma is a public health epidemic and continues to be the leading cause of death in children, adolescents and young adults in the United States.¹ Shock state is associated with poor outcomes in injured children,² and its diagnosis is often delayed due to preserved hemodynamics despite significant blood loss and hemostatic derangement. Although traumatic brain injury is the predominant source of mortality and morbidity, hemorrhage remains a common cause of preventable death in both adults and children.^{3–5} Furthermore, trauma-induced coagulopathy is common, and is associated with poor outcomes in injured pediatric patients.^{6–8} Injured children requiring massive transfusion have a greater than 50% mortality rate⁹ and died at a rate 2 to 3 times greater than adult patients with traumatic injury.¹⁰ Therefore, the early recognition of shock and aggressive hemostatic resuscitation are of critical importance in the management of severe pediatric trauma.

Low titer group O whole blood (LTOWB) is utilized as the initial resuscitative fluid in an increasing number of adult trauma and massive bleeding transfusion protocols.¹¹ LTOWB has an excellent safety profile in adults^{12–14} and some studies report improved survival in adult LTOWB recipients as compared to component therapy recipients.^{15–17} In children, transfusion of LTOWB has been shown to be feasible, efficient, and safe in small pediatric trauma cohorts.^{18–21} These studies are limited by the small number of subjects and modest transfusion volumes of LTOWB; to date, there has not been a pediatric study that reported improved survival after LTOWB transfusion. This study was performed to determine if the administration of LTOWB during the trauma resuscitation of pediatric patients requiring massive transfusion led to improved survival compared to a historical cohort of patients who received conventional components only.

MATERIALS AND METHODS

LTOWB Protocol

The methods for LTOWB collection and storage at this facility have been described in detail elsewhere.²¹ In brief, group O whole blood units were collected from male donors with low titers (<50) of both anti-A and anti-B, leukoreduced with a platelet-sparing filter and stored between 1°C and 6°C in the Emergency Department (ED) and in the hospital's blood bank. RhD-negative units are used to allow both male and female children to receive LTOWB while eliminating the risk of RhD alloimmunization. If the LTOWB was not used by storage day 14, it was recovered by the blood bank to be processed into a red blood cell (RBC) unit. When LTOWB first was made available for injured children at this institution, the following restrictions were in place: maximum permitted LTOWB volume was initially 20 mL/kg, and the patients also had to be age ≥3 years and

weigh ≥ 15 kg. Based upon safety data, the exclusion criteria for administration have been reduced over time; presently, this institution's guidelines permit the transfusion of up to 40 mL/kg of LTOWB in children age ≥ 1 year with signs of shock, after which conventional blood components are provided as needed based on the patient's clinical and laboratory findings.

Study Design and Data

This was an observational study using a prospectively maintained database of injured children presenting to UPMC Children's Hospital of Pittsburgh between January 2013 and December 2020. Injured children ages 0 to 17 years who required massive transfusion, defined as having received >40 mL/kg of blood products²² over the initial 24 hours of admission, were selected from the institutional trauma database. The total blood product volume included both LTOWB and conventional components. Children were excluded from analysis if they had pre-hospital cardiac arrest with death pronouncement on arrival to the trauma bay. Data extracted included demographics, injury mechanism, injury severity score (ISS), admission vital signs and Glasgow Coma Score (GCS), laboratory values including admission base deficit and international normalized ratio (INR), blood product volumes and time of administration, hemostatic adjuncts, morbidities, and mortality. Subjects were categorized by receipt of LTOWB as part of their hemostatic resuscitation (LTOWB group) versus receipt of conventional components only (component therapy group, CT). The patients in the LTOWB group all received LTOWB as their first resuscitative fluid and could also have received subsequent conventional components during their resuscitation. Those who received component therapy were either ineligible for LTOWB (by age or weight criteria) or presented before the initiation of the LTOWB program at this institution (2016). LTOWB volume was reported separately and not included in the component product volumes. The primary outcome was survival at 28-days post-injury. Secondary outcomes included 6-hour and 72-hour survival, acute kidney injury, intensive care unit (ICU) length of stay (LOS), hospital LOS, ventilator days, reported transfusion reactions,

venous thromboembolism and functional disability at discharge as defined by discharge to rehabilitation facility or score of "dependent" on functional independence measure (FIM) assessment. All research procedures conducted for this study were approved by the University of Pittsburgh Institutional Review Board.

Analysis

Data were summarized as counts, median (interquartile range), or percentage. Univariate analyses included Wilcoxon ranksum and Kruskal-Wallis testing for skewed continuous data and Chisquared or Fisher exact test for categorical data. A Kaplan-Meier analysis was used to calculate the probability of 6-hour, 72-hour, and 28-day survival as a function of time. Log rank testing for equality of survivor functions was then performed. To adjust for potential confounding factors, Cox proportional hazards regression analysis was then done using the outcomes of mortality at 6 hours, 72 hours, and 28 days. Variables included in the model generation were those with a $P < 0.2$ in univariate analyses (age, individual component, and total blood product transfusion volumes) and those selected *a priori* due to prior association with mortality in pediatric trauma (hypotension, base deficit, INR, ISS, injury mechanism, urgent surgery, admission Glasgow Coma Score). Variables were chosen for the final model using forward stepwise selection to minimize Bayesian Information Criteria (BIC). Collinear variables were excluded to minimize redundant information. All hypothesis tests were two-sided. Statistical analyses were conducted using STATA software, version 16 (StataCorp). Note that certain outcomes and other data on some of these patients has been published previously.^{19,20,21}

RESULTS

A total of 99 injured children received a massive transfusion in their first 24-hours after admission; 19 children were excluded from analysis because of death on arrival (12 in CT group, 7 in LTOWB group). This excluded cohort had

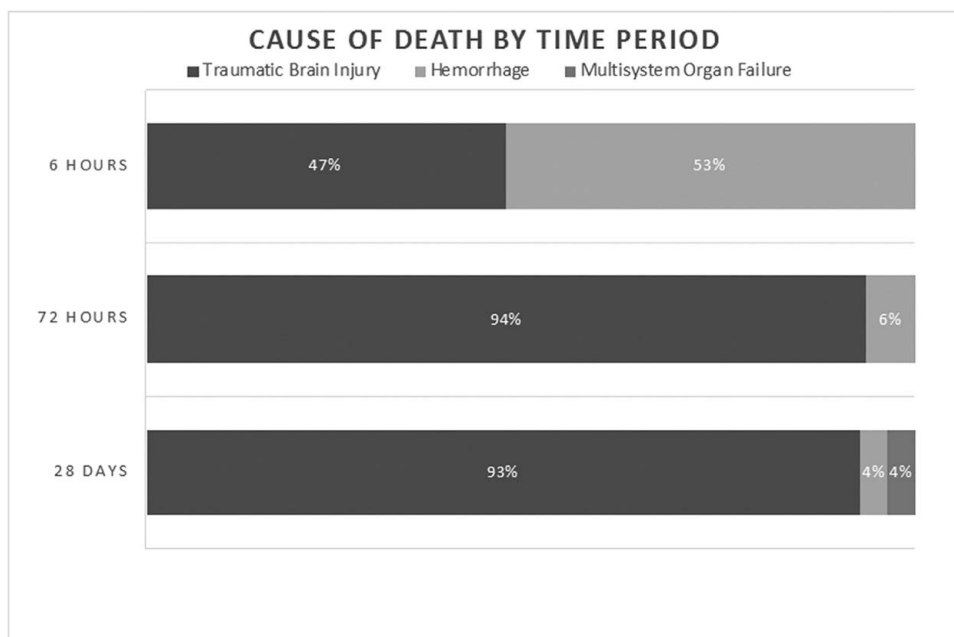


FIGURE 1. Cause of death by time period.

statistically similar age, sex, ISS, and mechanism to the overall analysis cohort. The remaining 80 patients included in the final analysis had median (IQR) age of 6 (2–12) years and median (IQR) ISS of 30 (21–38). The mortality rates at each time point for the whole cohort were: 6-hour mortality 19% (15/80), 72-hour mortality 39% (31/80), 28-day mortality 56% (45/80). No child died between 6 and 24 hours after injury, therefore 24-hour mortality was not analyzed further. Cause of death was attributable to brain injury in 76% (34/45), hemorrhage in 22% (10/45) and multisystem organ failure in 2% (1/45). Early death

(6 hours) was more likely to be caused by hemorrhage and later death (72 hours, 28 days) was more likely to be caused by brain injury ($P = 0.01$; Fig. 1).

Altogether, 33% (27/80) of patients received LTOWB as part of their initial hemostatic resuscitation. The LTOWB group was comparable to the component therapy group (Table 1) on baseline demographic and physiologic parameters except for older age and higher body weight, as expected due to age and weight restrictions that were in place at this institution for LTOWB use. There were no differences in admission laboratory

TABLE 1. Demographics and Characteristics of CT Group Versus LTOWB Group

	Component Therapy (n = 53)	Whole Blood (n = 27)	P
Demographics (n = 80)			
Age, y	5 (2–11)	8 (4–13)	0.03
Weight, kg	20 (9–45)	30 (16–52)	0.01
Sex			
Male	31 (58%)	14 (52%)	0.64
Female	22 (42%)	13 (48%)	
Race			
White	35 (74%)	21 (78%)	0.83
Black	8 (17%)	5 (18%)	
Other	3 (6%)	1 (4%)	
Injury characteristics (n = 80)			
Time from injury to hospital arrival, min	85 (59–169)	74 (36–160)	0.64
Mechanism of injury			
Blunt	42 (89%)	23 (85%)	0.71
Penetrating	5 (11%)	4 (15%)	
Injury Severity Score	33 (22–38)	27 (20–38)	0.85
Urgent SURGERY	24 (45%)	13 (48%)	0.64
Time to craniotomy, min	54 (35–60)	83 (49–92)	0.86
Admission characteristics (n = 80)			
Lowest Glasgow Coma Scale	3 (3–8)	3 (3–15)	0.40
Lowest temperature	36.3 (35.6–36.9)	35.8 (35.2–36.9)	0.29
Age-adjusted hypotension	12 (26%)	7 (26%)	1.00
Highest heart rate	132 (96–158)	121 (106–140)	0.33
Pre-resuscitation laboratory assays (n = 80)			
Hemoglobin	10 (8–12)	10 (8–12)	0.76
International normalized ratio	1.5 (1.3–2.1)	1.5 (1.3–2.2)	0.90
Platelet count	230 (165–325)	208 (145–289)	0.25
Base deficit	8 (5–13)	7 (3–13)	0.40
Thromboelastography (n = 50)			
ACT	128 (105–167)	121 (113–152)	1.00
R	0.9 (0.6–2.3)	0.8 (0.7–1.2)	0.77
K	1.7 (0.8–2.6)	2.1 (1.3–3.2)	0.51
MA	58.3 (38.8–62.2)	54.8 (46.1–62.8)	0.75
LY30	1.6 (0–6.7)	0.7 (0–3.9)	0.74
Blood products and adjuncts (n = 74)			
Total volume, mL/kg	61 (45–84)	50 (41–74)	0.06
Whole blood, mL/kg	-	20 (15–33)	-
Red blood cell volume, mL/kg	30 (20–60)	18 (12–25)	0.002
Plasma volume, mL/kg	30 (14–37)	15 (7–30)	0.02
Platelet volume, mL/kg	4 (0–10)	2 (0–5)	0.36
Cryoprecipitate (yes/no)	19 (37%)	9 (33%)	1.00
Plasma:red blood cell ratio	0.74 (0.32–1.33)	0.70 (0.42–1.18)	0.60
Platelet:red blood cell ratio	0.05 (0–0.25)	0.09 (0–0.21)	0.79
Tranexamic acid	6 (11%)	6 (22%)	0.33
Outcomes			
Time to death, h	46 (4–74)	120 (46–156)	0.03
Reported transfusion reaction (n = 80)	0 (0%)	0 (0%)	1.00
Acute kidney injury (n = 35 survivors)	3/20 (15%)	1/15 (7%)	0.63
Venous thromboembolism (n = 35 survivors)	7/20 (35%)	3/15 (20%)	0.47
Functional disability at discharge (n = 35 survivors)	11/20 (55%)	5/15 (33%)	0.32
Hospital length of stay (n = 35 survivors)	19 (12–29)	10 (5–20)	0.02
ICU length of stay (n = 35 survivors)	9 (4–18)	2 (1–7)	0.02
Ventilator days (n = 35 survivors)	6 (1–12)	1 (0–4)	0.05

Numeric values are median (IQR) when continuous and n (%) when discrete.

parameters including conventional coagulation tests and thromboelastography, nor in time from injury to hospital arrival (Table 1). Overall, 60% of LTOWB recipients received the maximum allowable dose of LTOWB based on restrictions in place at the time of transfusion. Regarding transfusion volumes, the CT group tended to receive greater volumes of RBC and plasma compared to the LTOWB group, as well as a greater total weight-adjusted transfusion volume although the difference for total volume approached statistical significance ($P = 0.06$; Table 1). The resuscitation ratio for the CT group did not differ significantly from that of the LTOWB group (Table 1). The causes of death did not differ significantly between LTOWB and CT groups ($P = 1.0$). The median time to death was significantly shorter for the CT group compared to the LTOWB group (Table 1).

On univariate analysis, survival was increased at 72 hours (log rank $p = 0.01$) in recipients of LTOWB compared to the CT group; 6-hour (log rank $P = 0.23$) and 28-day (log rank $P = 0.06$) mortality did not reach statistical significance (Fig. 2). After adjusting for age, ISS, total blood product volume transfused, admission base deficit, and admission INR in a Cox proportional hazard regression model, children who received LTOWB as part of their resuscitation had significantly decreased mortality at both 72 hours and 28 days posttrauma [adjusted odds ratio (AOR) 0.23, $P = 0.009$ and AOR 0.41, $P = 0.02$, respectively] (Table 2).

There were no significant baseline differences between survivors in terms of age, sex, injury severity score, or injury mechanism (all $P > 0.22$). Regarding secondary outcomes, on average, survivors in the LTOWB cohort had reduced hospital length of stay, ICU length of stay, and ventilator days compared to survivors in the CT cohort (Table 1). There were no significant

differences in any of the safety outcomes including transfusion reactions, acute kidney injury, or venous thromboembolism. There was also no difference in functional disability at discharge in survivors between the 2 groups (Table 1).

DISCUSSION

This is the first prospective observational pediatric study to show that LTOWB was independently associated with decreased 72-hour and 28-day mortality compared to component therapy alone when utilized in the hemostatic resuscitation of injured children who required a massive transfusion. Regarding other clinical outcomes, survivors in the LTOWB group had reduced hospital LOS, ICU LOS, and ventilator days compared to the CT group.

Adult trauma cohort studies have reported decreased mortality and reduction in post-ED blood product transfusion following the transfusion of LTOWB compared to conventional components,^{12–16,23} although the only randomized-controlled trial to date in an adult trauma cohort using modified (platelet-reduced) whole blood where the primary endpoint was blood product transfusion volumes did not show a mortality benefit.²⁴ Previous studies of LTOWB in pediatric trauma focused on the serologic safety and efficiency of its administration. In the first cohort of injured civilian children, LTOWB was associated with a faster time to the transfusion of at least 1 U of RBC, plasma, and platelets compared to the use of conventional components, without an increased risk of hemolysis.²¹ In an expanded cohort of pediatric patients with both medical and traumatic bleeding, there was no increased risk of hemolysis¹⁸ nor increased risk in adverse events such as organ failure or thromboembolic complications when LTOWB was transfused compared to patients

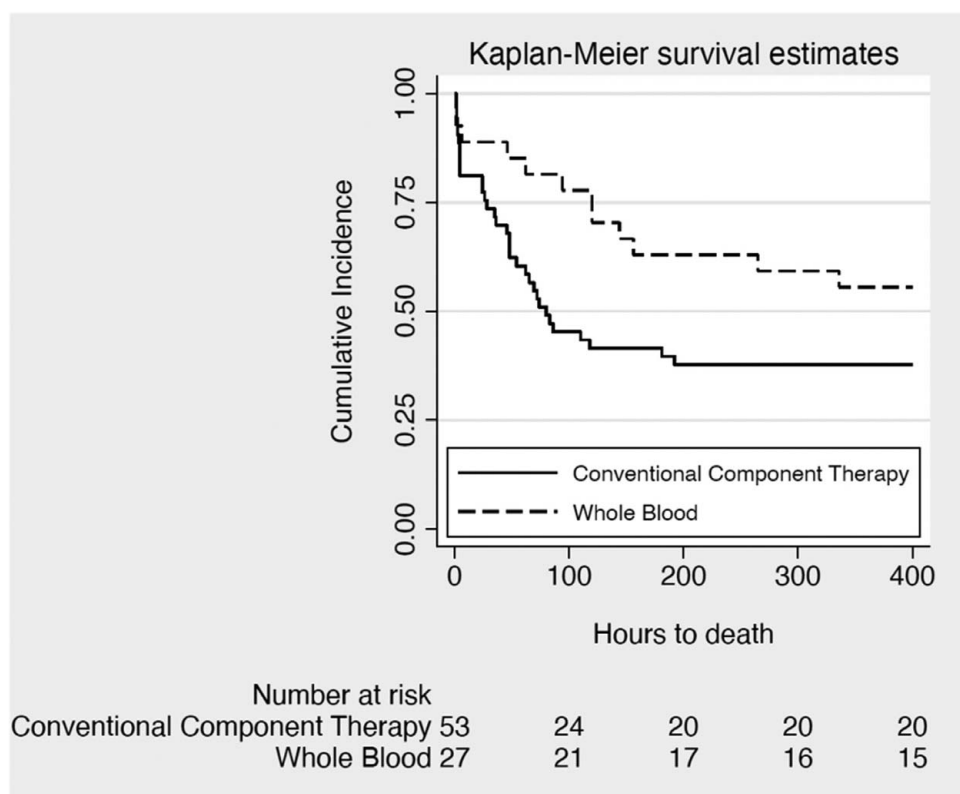


FIGURE 2. Children who received LTOWB as part of their resuscitation had significantly decreased mortality at both 72 hours and 28 days post-trauma (AOR 0.23, $P = 0.009$ and AOR 0.41, $P = 0.02$, respectively).

TABLE 2. Cox Hazard Regression Models for 72-Hour and 28-Day Mortality in Injured Children Requiring Massive Transfusion

	72-h Mortality			28-Day Mortality		
	Odds Ratio	95% Confidence Interval	P	Odds Ratio	95% Confidence Interval	P
LTOWB	0.23	0.08–0.70	0.009	0.41	0.23–0.98	0.02
Age, y	1.07	1.00–1.16	0.06	1.02	0.98–1.09	0.13
Total transfusion volume, mL/kg	1.01	1.01–1.02	<0.001	1.01	1.01–1.02	<0.001
Admission base deficit	1.11	1.06–1.17	<0.001	1.12	1.07–1.17	<0.001
Admission INR	1.30	1.06–1.58	<0.001	1.29	1.05–1.58	0.003
Injury Severity Score	1.02	1.00–1.05	0.10	1.02	0.99–1.06	0.06

who received conventional components.²⁵ This is concordant with safety data in adult trauma patients.^{12–16,23,26}

In a previous propensity analysis of injured children, recipients of LTOWB had faster resolution of shock, lower post-transfusion INR and fewer overall blood products transfused compared to patients who received conventional components,¹⁹ but no mortality benefit was seen. In a second pediatric cohort propensity analysis looking at adverse events after LTOWB transfusion, a reduction in ICU length of stay was seen in the LTOWB group as compared to the CT group, but again no mortality benefit was seen.²⁵ Another recent propensity analysis used the Trauma Quality Improvement Program (TQIP) database, which included patient data reported in the aforementioned studies in addition to other pediatric LTOWB recipients across several US institutions.²⁷ The median age in this study was 12, and children received median 15 mL/kg of LTOWB; in general, decreased blood product volumes were utilized and there was a reduction in ventilator days for recipients of LTOWB, but there was no difference in survival between LTOWB and CT groups.

This report is the first to show improved survival in a pediatric trauma cohort who received LTOWB. Two key factors may explain the mortality benefit seen in this report. First, the present cohort included only children who received a massive transfusion of blood products. Therefore, this study may have selected for a cohort who was more likely to benefit from early, balanced transfusion, as compared to children who require smaller volumes of transfusion during their initial resuscitation such as those who were included in previous studies. Second, survival could be attributable to the dose of LTOWB transfused. Previous work from this institution included children who received a smaller median volume of LTOWB, 15 mL/kg,^{19,21} a volume similar to what was administered to LTOWB recipients in the TQIP database study.²⁷ At that time, this institution's maximum permitted LTOWB volume was 20 mL/kg, and the patients also had to be age ≥ 3 years and weigh ≥ 15 kg. Based upon these safety data, the exclusion criteria for administration have been reduced over time, such that now a maximum volume of 40 mL/kg can be administered to children age > 1 year, and there is no longer a minimum weight criterion for receipt of LTOWB at this institution. This present cohort includes children who received LTOWB under previous restrictions; therefore, the median volume of LTOWB transfused in this cohort is less than the present maximum allowable volume. Furthermore, only 60% of children received the maximum allowable volume of LTOWB based on restrictions in place at the time of transfusion; this is likely multifactorial, due to clinician familiarity with the protocol, product availability, and patient weight. Nevertheless, as children receive greater weight-based dosing of LTOWB in their resuscitation, additional benefits, including improved survival, may be seen as demonstrated in this study.

Children with life-threatening bleeding have an exceptionally high mortality rate.^{9,10} In this study population, as

described in other pediatric cohorts, those requiring massive transfusion had a $> 50\%$ mortality rate. Importantly, the etiology of death was influenced by the timing of death; beyond 6 hours post-injury, death was principally due to traumatic brain injury. Children in cardiac arrest on presentation had injuries that were nonsurvivable and therefore were excluded from the analysis. Those who died within 6 hours of presentation tended to die of hemorrhage more often than those who died later in their admission; however, the low occurrence of early death precluded a robust analysis of the effect of LTOWB in this subgroup. Those who died within 72 hours of presentation predominantly had sustained severe TBI and death was attributable to their central nervous system injury; this subset of patients was not coagulopathic based on either conventional coagulation parameters or thromboelastography, yet saw a clear benefit from LTOWB administered early during resuscitation. This suggests that the mechanism by which LTOWB increases survival is not solely by volume expansion and treatment of hemorrhagic shock, an interpretation which is supported by prior work in adults showing that early resuscitation with plasma has a survival benefit,²⁸ especially for those with TBI.²⁹ Further study into the effects of LTOWB in resuscitation of children with severe TBI is warranted.

Limitations to the study include the relatively small number of subjects, as pediatric massive transfusion after trauma is relatively uncommon. Further, mortality within the first 6 hours after injury and mortality due to bleeding were also infrequent outcomes, which precluded a robust analysis of the impact of LTOWB on very early death and death due to hemorrhage. Future multicenter studies with a larger number of subjects should be undertaken to address these limitations. These data were drawn from a single center which may limit generalizability to other settings, including those where transfusion is less common. The groups were not randomized and therefore differences between groups may exist that were not accounted for, although the adjusted analysis included many potential important covariates. Furthermore, transfusion reactions, which are passively reported to the blood bank for more detailed investigation at this hospital, might have been missed due to underreporting; although it is unlikely, but still possible, that a major transfusion reaction such as an immediate hemolytic reaction or a severe allergic reaction with hypotension would go undetected or unreported, it is possible that reactions such as those involving urticaria or fevers were not captured in this study. As an increasing number of centers are utilizing LTOWB in children, ongoing accumulation of safety data will clarify the true incidence of these generally low-frequency events.

CONCLUSIONS

Administration of LTOWB as part of hemostatic resuscitation in injured children who required massive transfusion was

independently associated with improved 72-hour and 28-day survival; this may be due to reduction in death due to traumatic brain injury. LTOWB was found to be effective as the initial resuscitative fluid in this cohort of pediatric trauma patients; additional studies are needed to support this finding.

REFERENCES

- Drake SA, Holcomb JB, Yang Y, et al. Establishing a regional trauma preventable/potentially preventable death rate. *Ann Surg*. 2020;271:375–382.
- Leeper CM, McKenna C, Gaines BA. Too little too late: hypotension and blood transfusion in the trauma bay are independent predictors of death in injured children. *J Trauma Acute Care Surg*. 2018;85:674–678.
- Fox N, Goldenberg SP, Porter AJ. Pediatric mortality and preventable death at a mature trauma center. *J Emerg Med Crit Care*. 2018;4:4.
- Kwon AM, Garbett NC, Kloecker GH. Pooled preventable death rates in trauma patients: Meta analysis and systematic review since 1990. *Eur J Trauma Emerg Surg*. 2014;40:279–285.
- Davis JS, Satahoo SS, Butler FK, et al. An analysis of pre-hospital deaths: Who can we save? *J Trauma Acute Care Surg*. 2014;77:213–218.
- Leeper CM, Kutcher M, Nasr I, et al. Acute traumatic coagulopathy in a critically injured pediatric population: definition, trend over time, and outcomes. *J Trauma Acute Care Surg*. 2016;81:34–41.
- Leeper CM, Neal MD, McKenna C, et al. Abnormalities in fibrinolysis at the time of admission are associated with deep vein thrombosis, mortality, and disability in a pediatric trauma population. *J Trauma Acute Care Surg*. 2017;82:27–34.
- Liras IN, Caplan HW, Stensballe J, et al. Prevalence and impact of admission acute traumatic coagulopathy on treatment intensity, resource use, and mortality: an evaluation of 956 severely injured children and adolescents. *J Am Coll Surg*. 2017;224:625–632.
- Shroyer MC, Griffin RL, Mortellaro VE, et al. Massive transfusion in pediatric trauma: analysis of the National Trauma Databank. *J Surg Res*. 2017;208:166–172.
- Leonard JC, Josephson CD, Luther JF, et al. Life-threatening bleeding in children: a prospective observational study. *Crit Care Med*. 2021;49:1943–1954.
- Yazer MH, Spinella PC, Anto V, et al. Survey of group A plasma and low titer group O whole blood use in trauma resuscitation at adult civilian Level I trauma centers in the US. *Transfusion*. 2021;61:1757–1763.
- Sehult JN, Bahr M, Anto V, et al. Safety profile of uncrossmatched, cold-stored, low-titer, group O+ whole blood in civilian trauma patients. *Transfusion*. 2018;58:2280–2288.
- Sehult JN, Triulzi DJ, Alarcon LH, et al. Measurement of haemolysis markers following transfusion of uncrossmatched, low-titre, group O whole blood in civilian trauma patients: initial experience at a level I trauma centre. *Transfus Med*. 2017;27:30–35.
- Yazer MH, Jackson B, Sperry JL, et al. Initial safety and feasibility of coldstored uncrossmatched whole blood transfusion in civilian trauma patients. *J Trauma Acute Care Surg*. 2016;81:21–26.
- Williams J, Merutka N, Meyer D, et al. Safety profile and impact of low-titer group O whole blood for emergency use in trauma. *J Trauma Acute Care Surg*. 2020;88:87–93.
- Hanna K, Bible L, Chehab M, et al. Nationwide analysis of whole blood hemostatic resuscitation in civilian trauma. *J Trauma Acute Care Surg*. 2020;89:329–335.
- Shea SM, Staudt A, Thomas KA, et al. The use of low-titer group O whole blood is independently associated with improved survival compared to component therapy in adults with severe traumatic hemorrhage. *Transfusion*. 2020;60(suppl 3):S2–S9.
- Morgan KM, Yazer MH, Triulzi DJ, et al. Safety profile of low titer group O whole blood in pediatric patients with massive hemorrhage. *Transfusion*. 2021;61(suppl 1):S8–S14.
- Leeper CM, Yazer MH, Triulzi DJ, et al. Whole blood is superior to component transfusion for injured children: a propensity matched analysis. *Ann Surg*. 2020;272:590–594.
- Leeper CM, Yazer MH, Cladis FP, et al. Cold-stored whole blood platelet function is preserved in injured children with hemorrhagic shock. *J Trauma Acute Care Surg*. 2019;87:49–53.
- Leeper CM, Yazer MH, Cladis FP, et al. Use of uncrossmatched cold-stored whole blood in injured children with hemorrhagic shock. *JAMA Pediatr*. 2018;172:491–492.
- Neff LP, Cannon JW, Morrison JJ, et al. Clearly defining pediatric massive transfusion: cutting through the fog and friction with combat data. *J Trauma Acute Care Surg*. 2015;78:22–28. discussion 28–29.
- Duchesne J, Smith A, Lawicki S, et al. Single institution trial comparing whole blood vs balanced component therapy: 50 years later. *J Am Coll Surg*. 2021;232:433–442.
- Cotton BA, Podbielski J, Camp E, et al. A randomized controlled pilot trial of modified whole blood versus component therapy in severely injured patients requiring large volume transfusions. *Ann Surg*. 2013;258:527–532. discussion 532–523.
- Leeper CM, Yazer MH, Morgan KM, et al. Adverse events after low titer group O whole blood versus component product transfusion in pediatric trauma patients: a propensity-matched cohort study. *Transfusion*. 2021;61:2621–2628.
- Yazer MH, Freeman FA, Harrold IM, et al. Injured recipients of low titer group O whole blood have similar clinical outcomes compared to recipients of conventional component therapy: a single-center, retrospective study. *Transfusion*. 2021;61:1710–1720.
- Anand T, Obaid O, Nelson A, et al. Whole blood hemostatic resuscitation in pediatric trauma: a nationwide propensity-matched analysis. *J Trauma Acute Care Surg*. 2021;91:573–578.
- Sperry JL, Guyette FX, Brown JB, et al. Prehospital plasma during air medical transport in trauma patients at risk for hemorrhagic shock. *N Engl J Med*. 2018;379:315–326.
- Gruen DS, Guyette FX, Brown JB, et al. Association of prehospital plasma with survival in patients with traumatic brain injury: a secondary analysis of the PAMPer cluster randomized clinical trial. *JAMA Netw Open*. 2020;3:e2016869.